INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

BUSINESS CONTINUITY AND DISASTER RECOVERY

GRC205 - Incident Management and Business Continuity

Week 21 Assignment

Northbridge Skills Institute

Business Impact Analysis, Recovery Objectives,
Strategy Selection and Coordinated Continuity Plan

Student Name: **SAMUEL ADEOLUWA OKUNRIBIDO**
Student ID: **GCR2512878**
Official Email: **gcr2512878@students.icdfa.edu.ng**
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Executive Summary

This report presents a complete Business Continuity and Disaster Recovery (BC/DR) response for the fictional Northbridge Skills Institute following a sudden site outage caused by flooding and power failure at 08:00 on Day 0. The main campus is expected to remain inaccessible for 72 hours. The analysis covers Business Impact Analysis (BIA), recovery objectives, strategy selection within a fixed annual budget, coordinated continuity and recovery planning, schedule validation, tabletop exercise findings, and programme governance.

Five critical functions were assessed. Teaching and assessment (F1) carries the highest priority because of an 11:00 examination. Estimated disruption costs reach NGN 1,200,000 after eight hours and NGN 3,360,000 after twenty-four hours when no recovery measures are applied. Recovery Time Objectives (RTOs), Recovery Point Objectives (RPOs) and Maximum Tolerable Downtimes (MTDs) were set for each function, with deliberate buffers between technical recovery and business tolerance limits.

A warm-site strategy with an annual cost of NGN 2,100,000 (including mandatory independent backups and training) is recommended. It meets the majority of recovery targets while remaining inside the NGN 3,000,000 planning ceiling. Baseline warm-site timings show assessment service available at 11:45 and full functional recovery by 12:30, confirming that temporary examination arrangements remain necessary. A tabletop exercise exposed residual risks around backup integrity, workstation capacity and extended outage duration; five corrective actions were defined with owners and measurable evidence.

The report concludes that a structured, tested BC/DR programme aligned with NIST SP 800-34 and ISO 22301 principles can keep learner impact within acceptable bounds, provided residual risks are formally accepted and continuous improvement is maintained.

Lab Objectives

The assignment addresses the following learning outcomes:

I. Conduct a Business Impact Analysis and prioritise critical functions.
II. Justify recovery time and recovery point objectives using business requirements.
III. Compare alternate sites and backup strategies against cost and capability.
IV. Coordinate continuity, disaster recovery and incident response responsibilities.
V. Test recovery assumptions and convert findings into measurable improvements.

Tools and Resources Used

The following resources underpinned the analysis:

I. ICDFA GRC205 Week 21 teaching deck (slides 1-15) - primary instructional material on BIA, RTO/RPO/MTD, strategy options and governance.
II. NIST Special Publication 800-34 Revision 1 - Contingency Planning Guide for Federal Information Systems (Swanson et al., 2010).
III. ISO 22301:2019 - Security and resilience — Business continuity management systems — Requirements, together with the 2024 climate-action amendment.
IV. Supplied scenario data: function impact rates, backup catalogue, warm-site drill timings, package costs and resource constraints.
V. Standard office productivity tools for calculation, scheduling and document production.

Analysis and Findings

The BIA established a clear priority order driven by hard deadlines, while the technical dependency chain showed that shared infrastructure (network, AUTH01, DB01) constrains the speed at which business priority can be realised.

The 45-minute overrun on the F1 RTO is therefore structural rather than a scheduling error; continuity workarounds are essential even when recovery proceeds according to plan.

Checkpoint analysis confirmed that the existing 07:00 incremental meets the most demanding RPO, yet the older full backups and AUTH01 snapshot leave material gaps. The redesigned backup schedule with hourly incrementals and an independent offline copy closes those gaps and satisfies the 3-2-1 principle.

Cost-benefit comparison showed that the Warm package delivers the majority of required recovery capability at a sustainable annual cost. The tabletop exercise demonstrated that residual risks—extended outage, reduced workstation capacity, backup integrity failure and reconciliation discrepancies—can be managed provided pre-agreed decision rights and corrective actions are in place.

Overall, the institute can keep learner and staff impact within acceptable bounds by combining a tested warm-site capability, robust temporary procedures, and disciplined governance. The principal remaining exposure is the integrity and independence of the backup chain; addressing that exposure is the highest-priority corrective action.

11. Main Challenges

1. Shared technical foundations create a single critical path that cannot be shortened by additional parallel effort once the two-technician limit is reached.
2. The 11:00 examination falls inside the technical recovery window, forcing reliance on temporary assessment arrangements whose integrity and accessibility must be carefully controlled.
3. Backup administrator identity is currently shared with production, undermining both security and recoverability assumptions.
4. Alternate-facility workstation capacity is finite and proved fragile under the tabletop inject; surge arrangements are not yet contractually guaranteed.
5. Climate-related flood risk is acknowledged but not yet fully embedded in the periodic review cycle of the alternate site.

12. Conclusions and Recommendations

The analysis demonstrates that a structured BC/DR programme, grounded in a thorough BIA and realistic recovery objectives, can protect critical educational services within the constraints of a modest annual budget. The Warm package, supported by an improved backup regime and tested temporary procedures, offers the best balance of capability, cost and residual risk.

Principal recommendations:

6. Adopt the Warm strategy package and fund the mandatory independent-backup and training additions.
7. Implement the redesigned backup schedule, including separation of administrator identities and 30-minute (or continuous) DB01 incrementals, within 45 days.
8. Formalise surge capacity and multi-day occupancy arrangements with the alternate facility.
9. Execute the five corrective actions arising from the tabletop, track them through the KPI dashboard, and report closure rates monthly.
10. Embed climate-related flood review into the annual plan-maintenance cycle and reassess alternate-site resilience accordingly.

With these measures in place, Northbridge Skills Institute will be positioned to maintain acceptable service levels for learners and staff under the scenarios examined in this report.

Task 1 - Business Impact Analysis

Five-Function BIA

The Business Impact Analysis covers the five critical functions of Northbridge Skills Institute. Impact figures treat the stated hourly rates as separate, non-overlapping incremental costs applied for the first 24 hours. The one-off examination rescheduling cost of NGN 120,000 is added once when the 11:00 trigger is met. All values are exercise estimates only.

F1 - Teaching and Assessment (LMS01 and DB01)

I. Owner role: Academic Operations Manager
II. Users affected: 1,800 learners (immediate focus on 120 examination candidates); academic staff
III. Critical resources: LMS01, DB01, AUTH01, recovery network, examination workstations or approved personal devices, invigilators, secure assessment environment
IV. Dependencies: Recovery network, AUTH01, DB01, LMS01; staff availability; secure access; communication channels; Microsoft 365 for identity and email fallback
V. Impact over time: High from 08:00. Examination integrity and learner access are time-critical. Failure to provide an acceptable service by 11:00 triggers an immediate one-off rescheduling cost of NGN 120,000 in addition to the ongoing hourly rate.
VI. Minimum acceptable service: Delivery of the scheduled examination to all 120 candidates by 11:00 using a combination of the 25 alternate-site workstations (priority for the 40 candidates who require centre machines) and candidate-owned devices under invigilator supervision. Manual paper-based assessment is acceptable only with prior Academic Board approval, dual invigilation, secure collection of scripts, and later electronic reconciliation. Assessment integrity, accessibility adjustments and identity verification must be maintained.
VII. Proposed recovery priority: 1 (highest)

F2 - Fees and Finance (FIN01 and DB01)

I. Owner role: Finance Manager
II. Users affected: Learners due to confirm fees; finance staff; external payment provider (available)
III. Critical resources: FIN01, DB01, AUTH01, recovery network, temporary receipt and wallet logging capability
IV. Dependencies: Recovery network, AUTH01, DB01, FIN01; staff; secure access; external payment gateway
V. Impact over time: Moderate-to-high. Fee confirmation deadline is 16:00 on Day 0. Local receipts and wallet records cannot be updated until systems are restored, creating reconciliation risk.
VI. Minimum acceptable service: Acceptance of payments through the external provider with manual logging of transactions on approved paper or spreadsheet forms. Each entry requires dual finance-staff sign-off, sequential numbering and secure storage pending electronic reconciliation. No automatic wallet credit is permitted until system restoration and validation.
VII. Proposed recovery priority: 2

F3 - Admissions and Registration (REG01 and DB01)

I. Owner role: Admissions Officer
II. Users affected: Prospective and current learners seeking registration changes; admissions staff
III. Critical resources: REG01, DB01, AUTH01, recovery network, temporary application logging
IV. Dependencies: Recovery network, AUTH01, DB01, REG01; staff; secure access
V. Impact over time: Moderate. Admissions close at 18:00 on Day 0. New applications and registration changes cannot be entered into the local system.
VI. Minimum acceptable service: Manual capture of applications and change requests on controlled forms with unique reference numbers, dual review and temporary holding in a secure location. Entries are batch-loaded and validated after system restoration. No final registration confirmation is issued until electronic records are complete.
VII. Proposed recovery priority: 3

F4 - Staff Payroll (HR01 and DB01)

I. Owner role: Human Resources Manager
II. Users affected: 90 staff; payroll and bank submission process
III. Critical resources: HR01, DB01, AUTH01, recovery network, bank submission channel
IV. Dependencies: Recovery network, AUTH01, DB01, HR01; staff; secure access

V. Impact over time: Lower immediate impact on Day 0, rising sharply toward the 12:00 Day 1 bank submission deadline. Incorrect or duplicate instructions would require manual correction and could affect staff morale and compliance.
VI. Minimum acceptable service: Preparation of payroll instructions from the last known good data set with manual verification against staff records. Dual HR and Finance sign-off is mandatory before any bank file is generated. No submission occurs until the data have been cross-checked.
VII. Proposed recovery priority: 4

F5 - Records and Certificates (FS01)

I. Owner role: Registry Manager
II. Users affected: Learners requesting certificates or records; external verification bodies
III. Critical resources: FS01, AUTH01, recovery network, secure file storage
IV. Dependencies: Recovery network, AUTH01, FS01; staff; access controls
V. Impact over time: Lower immediate financial impact. The internal 48-hour service commitment remains in force. Integrity and access restrictions are non-negotiable.
VI. Minimum acceptable service: Manual logging of requests with unique reference numbers. Fulfilment is deferred until FS01 is restored and integrity verified, unless an urgent external verification is approved by the Registry Manager with dual control. No certificates are issued from unverified copies.
VII. Proposed recovery priority: 5

People, premises, suppliers and information needs have been incorporated for each function: two technicians for technical recovery, business coordinators for temporary services and learner support, the alternate facility (25 workstations available from 10:00), Microsoft 365 for identity and email, and the external payment provider.

1.2 Impact Calculations (8-hour and 24-hour)

Disruption begins at 08:00. Hourly rates are applied continuously. The examination rescheduling cost is incurred once at 11:00 because no recovery measures are assumed.

Table 1. Eight-hour disruption cost (08:00-16:00)

Function	Hours	Hourly rate (NGN)	Sub-total (NGN)	One-off (NGN)	Total (NGN)
F1	8	60,000	480,000	120,000	600,000
F2	8	40,000	320,000	-	320,000
F3	8	20,000	160,000	-	160,000
F4	8	10,000	80,000	-	80,000
F5	8	5,000	40,000	-	40,000
Combined					**1,200,000**

Table 2. Twenty-four-hour disruption cost (08:00 Day 0 - 08:00 Day 1)

Function	Hours	Hourly rate (NGN)	Sub-total (NGN)	One-off (NGN)	Total (NGN)
F1	24	60,000	1,440,000	120,000	1,560,000
F2	24	40,000	960,000	-	960,000
F3	24	20,000	480,000	-	480,000
F4	24	10,000	240,000	-	240,000
F5	24	5,000	120,000	-	120,000
Combined					**3,360,000**

Two limitations of these estimates must be noted. First, the figures treat hourly costs as linear and independent; cascading effects (for example, delayed fee confirmation affecting subsequent registration activity) are not captured. Second, the estimates exclude non-financial consequences and assume constant demand. Actual learner behaviour, staff overtime and supplier response times would alter the totals.

Financial impact must be distinguished from learner welfare, reputation and data integrity. The calculated NGN figures represent direct incremental disruption costs only. Learner welfare is affected by examination integrity, accessibility and timely fee and registration processing. Reputation risk arises from any public perception of disorganisation or compromised assessment standards. Data integrity concerns the accuracy of restored records,

the risk of dual entries during manual workarounds, and the preservation of access controls on certificates and personal information. These dimensions cannot be expressed solely in the hourly rates supplied.

1.3 Priority Sequence and Dependency Risks

The 11:00 examination creates an absolute business priority for F1. The subsequent deadlines (fee confirmation at 16:00, admissions closure at 18:00, and payroll bank submission at 12:00 on Day 1) establish a clear business ranking: F1 -> F2 -> F3 -> F4 -> F5.

This business priority is distinct from the technical restoration order imposed by shared infrastructure. All functions F1-F4 require the recovery network, AUTH01 and DB01 before their individual applications can be restored. F5 requires the recovery network, AUTH01 and FS01. Consequently, the technical sequence must begin with network recovery, followed by AUTH01, security and backup checks, and then DB01 (or FS01 in parallel once AUTH01 is ready). Only after these shared foundations are complete can the application-specific restorations proceed according to business priority.

Three important dependency risks were identified:

1. Single point of failure in the shared recovery network and AUTH01 - any delay or failure here blocks every subsequent function.
2. Shared administrator identity between production and network backup administration - this creates both a security exposure and a potential bottleneck if the same individual is required for multiple concurrent recovery tasks.
3. Limited technician capacity (only two technicians) combined with non-splittable recovery actions - parallel work is possible only when prerequisites are satisfied and both technicians are free, which constrains the overall recovery timeline even when business priority would prefer earlier availability of a specific function.

Task 2 - Recovery Objectives and Backup Design

2.1 Definitions and Proposed Targets

Recovery Time Objective (RTO) is the maximum acceptable time from the point of disruption until a defined service level is restored and available for use. Recovery Point Objective (RPO) is the maximum acceptable age of the data that can be restored; it defines the tolerable data-loss window measured backwards from the moment of disruption. Maximum Tolerable Downtime (MTD) is the total period of disruption the organisation can endure before the impact becomes unacceptable; it is the outer business limit within which RTO must sit.

A deliberate buffer is retained between technical recovery (RTO) and MTD so that validation, business acceptance and any residual manual reconciliation can still occur inside the tolerable window.

Table 3. Recovery objective matrix

Function	RTO	RPO	MTD	Service-level declaration	Justification / buffer
F1 Teaching	3 h	1 h	5 h	LMS01 available; exam can be delivered	Exam at 11:00; 2 h buffer for setup
F2 Fees	6 h	2 h	10 h	FIN01 or dual-controlled manual log	Deadline 16:00; buffer for recon.
F3 Admissions	8 h	4 h	12 h	REG01 or controlled manual capture	Close 18:00; buffer for review
F4 Payroll	20 h	4 h	28 h	HR01 available; payroll validated	Bank 12:00 Day 1; dual sign-off buffer
F5 Records	24 h	12 h	48 h	FS01 restored; integrity verified	Matches 48 h commitment

2.2 Checkpoint Age and Data-Loss Windows at 08:00

Table 4. Checkpoint analysis

Item	Checkpoint	Age at 08:00	Functions	Loss window	vs proposed RPO
B01 DB01 full	23:00 Day -1	9 hours	F1-F4	Up to 9 h	Exceeds all
B02 Incremental	07:00 Day 0	1 hour	F1-F4	1 hour	Meets F1; OK F2-F4
B03 AUTH01	00:00 Day 0	8 hours	All	8 hours	Exceeds most
B04 FS01 full	23:00 Day -1	9 hours	F5	9 hours	Within F5 (12 h)

The 07:00 incremental is the latest recoverable point for business data. Temporary records created after that time can be reconciled later provided dual control and sequential numbering are applied. Changes that existed only in volatile memory or unbacked application caches cannot be assumed recoverable.

2.3 Backup Schedule Design

The schedule closes the identified RPO gaps and applies the 3-2-1 principle: three copies of data, on two different media or storage technologies, with one copy offline or immutable and geographically separated, under access controls independent of production administrators.

Table 5. Detailed backup schedule

Type	Frequency & timing	Retention	Media / location	Scope & chain	Controls
DB01 Full	Daily 23:00	14 d onsite 30 d offsite	Encrypted disk + object storage + offline	Base for all incrementals	Separate admin ID; dual-key encryption
DB01 Incr.	Every 60 min (or CDC)	7 days rolling	Same dual media as full	Full + unbroken chain to point	Checksum after each write; 15 min alert
AUTH01 Snapshot	Every 4 hours	7 days	Snapshot repo + weekly offline	Directory & config state	Directory admin only; no prod write access
FS01 Full	Daily 23:00	30 d offsite 90 d archive	Disk + object + monthly offline	Full file-level; ACLs preserved	Hash at backup time
Offline / immutable	Weekly Sat 02:00	12 weeks	Air-gapped media at remote secure site	Complete weekly set	Physical log; dual-control retrieval

Restoration of DB01 always requires the full backup plus the unbroken incremental chain to the chosen recovery point. Any break in the chain forces fall-back to the previous full and increases the data-loss window.

2.4 Recoverability Assurance

Offsite storage or a live replica alone does not establish recoverability. A replica may itself be corrupted or encrypted by the same event that affects production; an offsite copy whose access credentials are shared with production administrators can be deleted or altered by the same compromised identity. Integrity, access independence and usability are verified through cryptographic checksum comparison at write and before restore, quarterly isolated restoration tests with application-level validation, annual offline-copy retrieval tests under dual control, and documented separation of backup administrator identities from production administrators. Any assumption not tested in the last 90 days is labelled “unverified” in the recovery runbook and treated as a residual risk requiring management acceptance.

2.5 Task 2 Summary

Recovery objectives were defined and justified for each function, with explicit buffers between RTO and MTD. Checkpoint ages at the 08:00 disruption were calculated and mapped against the proposed RPOs, revealing that the 07:00 incremental meets the most stringent (F1) target while older full backups and the AUTH01 snapshot exceed several RPO values. A detailed backup schedule was designed that closes these gaps: hourly (or continuous) DB01 incrementals, four-hourly AUTH01 snapshots, daily fulls, and a weekly independent offline/immutable copy. The schedule fully implements the 3-2-1 principle and specifies the exact restore chain required for DB01. The

distinction between mere offsite presence and genuine recoverability was explained, together with concrete integrity, access-independence and usability checks.

Task 3 - Recovery Strategy and Budget

3.1 Comparison of Strategy Packages

The three packages are evaluated against the recovery objectives established in Task 2, staffing constraints, geographic and supplier considerations, and the residual risk of propagating corruption or malicious change. Quoted recovery windows remain planning estimates that require validation through testing.

Table 6. Strategy option comparison

Criterion	Cold	Warm	Hot
Base cost (NGN)	600,000	1,500,000	2,400,000
First critical service	24-48 hours	3-5 hours	≈ 1 hour
Data-loss window	Up to 24 hours	1-6 hours	Up to 15 min
F1 RTO/RPO fit	Does not meet	Meets RTO; RPO edge	Meets both
Corruption propagation risk	Lowest	Moderate	Highest
Within 3 M limit after additions	Yes	Yes	Yes (at limit)

3.2 Complete Annual Cost Calculation

Mandatory additions for every package: independent protected backups NGN 400,000 and training/tests NGN 200,000.

Table 7. Itemised annual budget

Package	Base (NGN)	Backups (NGN)	Training (NGN)	Total (NGN)
Cold	600,000	400,000	200,000	1,200,000
Warm	1,500,000	400,000	200,000	2,100,000
Hot	2,400,000	400,000	200,000	3,000,000

3.3 Recommendation and Residual Risks

The Warm package (total annual cost NGN 2,100,000) is recommended. It is the only option that simultaneously satisfies the majority of the RTOs established in Task 2 (particularly the critical 3-hour target for F1) while remaining comfortably inside the budget and preserving modest contingency. The Cold package fails the examination-driven RTO. The Hot package meets every numerical objective but exhausts the entire budget and elevates the risk that logical corruption could be replicated before detection.

Residual risks include a possible data-loss exposure of several hours for the tightest RPO, continued dependence on the two-technician model, supplier and geographic risk at the alternate facility, and the possibility that a sophisticated integrity failure could still affect warm-site replicas. These risks must be formally accepted by the Academic Operations Manager and the Finance Manager, with the BC/DR Coordinator retaining accountability for monitoring and escalation.

3.4 Comparison with Disruption Costs

The 8-hour disruption cost is NGN 1,200,000 and the 24-hour cost is NGN 3,360,000. The recommended Warm package annual outlay of NGN 2,100,000 is therefore lower than a single 24-hour outage. A single-outage estimate cannot by itself establish annual return on investment; outage frequency, secondary costs and the probability of concurrent failures are also required. Evidence needed before expenditure approval includes documented results of at least two successful recovery exercises, independent verification of backup integrity, written confirmation of alternate-facility capacity, and clear contractual recovery commitments with remedies for non-performance.

3.5 Task 3 Summary

The Cold, Warm and Hot packages were compared against RTO/RPO targets, staffing limits, geography, supplier dependency and corruption-propagation risk. Complete annual costs inclusive of the mandatory additions were

calculated; all three options remain inside the NGN 3,000,000 ceiling. The Warm package (NGN 2,100,000) is recommended as the balanced choice. Residual risks were identified and assigned for formal acceptance. The single-outage cost figures were contrasted with the annual investment, and the additional evidence needed before approving expenditure was specified.

Task 4 - Business Continuity and Disaster Recovery Plan

4.1 Coordinated Plan - Scope, Activation and Governance

Scope covers the coordinated response to unavailability of the main site and local systems for up to 72 hours (extendable). It addresses continuity of the five critical functions, technical recovery at the alternate facility, temporary business workarounds, stakeholder communication and controlled return to the primary site. Microsoft 365 identity and email remain available and serve as the primary communication channel.

Activation occurs immediately when the main site is declared inaccessible. Automatic escalation applies if the outage is projected to exceed four hours or if any critical deadline is at risk. Incident-response involvement is triggered by evidence of compromise or data-integrity failure. The primary decision maker is the BC/DR Coordinator; the deputy is the Academic Operations Manager. Situation reports are issued at activation, at each critical deadline, at the end of each operational day and on any material change.

4.2 RACI Matrix

Table 8. RACI matrix (A = Accountable, R = Responsible, C = Consulted, I = Informed)

Activity	Coord.	Acad Ops	Finance	HR	Registry	Tech/Fac
Activation	A/R	C	C	I	I	R / C
Temp assessment (F1)	A	R	I	I	C	C
Backup validation	A	I	I	I	I	R
System restoration	A	C	C	C	C	R
Stakeholder comms	A/R	R	C	C	C	I
Return to primary site	A	R	R	C	R	R

4.3 Temporary Procedures for the 72-Hour Outage

F1: Priority use of the 25 alternate-site workstations is given to the 40 candidates who require centre machines; the remaining 80 use their own devices under invigilation. Manual paper scripts require Academic Board approval, dual invigilation and later reconciliation.

F2: Payments continue via the external provider with dual-controlled sequential logging.

F3: Applications are captured on numbered forms, dual-reviewed and batch-loaded after restoration.

F4: Payroll is prepared from last known good data with dual HR/Finance sign-off.

F5: Requests are logged; fulfilment is deferred until FS01 integrity is verified except for urgent dual-controlled external verifications. All manual records are retained for audit; privacy and access restrictions are maintained.

4.4 DR Runbook

Ordered actions:

(1) establish recovery network;

(2) restore AUTH01;

(3) security and backup checks;

(4) restore DB01 (full + incremental chain);

(5-8) restore LMS01, FIN01, REG01, HR01;

(9) restore FS01;

(10) business acceptance;

(11) maintain rollback capability. Each step has an owner, completion check and contingency. Technical restoration is distinct from resumption of acceptable business service; business acceptance is the formal declaration point.

4.5 Communication and Return

A stakeholder matrix defines channels, frequency and owners for learners, staff, senior leadership, the payment provider and the alternate facility. An initial learner notice informs candidates of the outage, examination arrangements, fee and registration guidance, and the support channel. Any indication of possible compromise immediately engages the incident-response team. Return to the primary site is authorised only after facilities clearance, successful restoration and business validation of all five functions, controlled transfer of temporary records, and formal sign-off.

4.6 Task 4 Summary

A coordinated BC/DR plan was defined covering scope, activation thresholds, decision rights, escalation and team interaction. A RACI matrix assigns clear accountability for six key activities. Temporary procedures for each of the five functions address the 25-workstation constraint, approvals, integrity, privacy and later reconciliation. A sequenced DR runbook with owners, completion checks and contingencies distinguishes technical restoration from business acceptance. A stakeholder communication matrix and initial learner notice were prepared, together with criteria for engaging incident response and for controlled return.

Task 5 - Recovery Schedule and Exercise

5.1 Baseline Warm-Site Recovery Schedule

Two technicians are available. Actions cannot be split. Parallel work is permitted only when all prerequisites are complete and both technicians are free. Start time is 08:00.

Table 9. Recovery schedule

Action	Min	Prerequisite	Tech	Start	Finish
Recovery network	30	None	Tech 1	08:00	08:30
AUTH01 restoration	30	Recovery network	Tech 1	08:30	09:00
Security & backup checks	30	AUTH01	Tech 2	09:00	09:30
DB01 restoration	90	Security checks	Tech 1	09:30	11:00
FS01 restoration	60	AUTH01 + checks	Tech 2	09:30	10:30
LMS01 restoration	45	DB01	Tech 1	11:00	11:45
FIN01 restoration	30	DB01	Tech 2	11:00	11:30
REG01 restoration	30	DB01	Tech 2	11:30	12:00
HR01 restoration	45	DB01	Tech 1	11:45	12:30

Earliest assessment service (LMS01): 11:45. Feasible completion of all five functions: 12:30. Critical dependency path controlling assessment finish time: Recovery network -> AUTH01 -> Security and backup checks -> DB01 -> LMS01 (225 minutes).

5.2 Elapsed Times versus Objectives

LMS01 exceeds its 3-hour RTO by 45 minutes, confirming that the temporary examination procedure remains necessary. All other functions meet their RTO, MTD and RPO targets under the baseline model. These are modelled results, not a live test.

5.3 Tabletop Exercise

A 45-minute tabletop addressed four injects: site closure extended to five days; only 15 workstations available; DB01 incremental fails integrity check together with an unexpected administrator logon; payment-provider transactions absent from restored finance records. Decisions, owners, rationale and service impact were recorded for each inject. Five prioritised corrective actions were defined with owners, target dates and measurable completion evidence

(independent backup identity, 30-minute incrementals, surge workstation capacity, extended-outage playbook, and formal payment reconciliation procedure).

5.4 Task 5 Summary

A feasible two-technician recovery schedule was constructed. Assessment service becomes available at 11:45; all five functions are restored by 12:30. The controlling dependency path was identified. Elapsed times were compared with objectives; LMS01 exceeds its RTO, confirming the need for continuity workarounds. A tabletop exercise produced reasoned responses to all four injects and five measurable corrective actions.

Task 6 - Governance and Performance Measures

6.1 Ownership, Testing and Maintenance

Plan owner is the BC/DR Coordinator; final approval rests with Senior Leadership. A risk-based testing calendar specifies full recovery exercises twice yearly, tabletops quarterly, monthly backup restoration tests, monthly contact-list verification, and annual plan review (or after material change). Version control uses sequential numbering and a change log. Role-specific training is required within 30 days of appointment or plan update, with annual refreshers.

6.2 Framework Mapping

Four outputs align with NIST SP 800-34 steps (policy, BIA, strategy selection, plan development and testing). Four align with ISO 22301 themes (leadership, BIA/risk, strategies and solutions, exercise programme). One relevant climate consideration is the increasing frequency of extreme rainfall; the alternate site's flood resilience and power redundancy should be re-evaluated periodically under the ISO 22301 climate-action amendments.

6.3 Key Performance Indicators

Four KPIs were defined: recovery performance (≥ 90 % of functions within RTO), restoration-test coverage (100 % of critical sets per quarter), corrective-action closure (≥ 85 % on time), and training completion (100 % within 30 days / annual refresher). Each has formula, data source, target, reporting frequency and accountable owner.

6.4 Maturity and Improvement Target

Current maturity is assessed as Managed-to-Defined. The twelve-month target is a solid Defined level characterised by two successful full exercises, four tabletops, sustained KPI performance, operational independent backup identity and 30-minute incrementals, and a completed climate-related flood review.

6.5 Task 6 Summary

Plan ownership, a risk-based testing calendar, version control and training requirements were defined. Framework mappings to NIST SP 800-34 and ISO 22301 were provided, together with one climate consideration. Four measurable KPIs and a realistic twelve-month maturity target were set.

13. References

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14. Acknowledgement of Assistance

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Student Name: Samuel Adeoluwa Okunribido

Student ID: GCR2512878